

55. Trusted Secure IP (TSIP)

This MCU incorporates a Trusted Secure IP module to provide security functions. The module consists of an access management circuit, encryption engine, and random number generator. In combination with the Trusted Secure IP library, the Trusted Secure IP can prevent eavesdropping (confidentiality), falsification of information (integrity), and impersonation (authenticity).

Key information to be used in encrypting and decrypting data is only stored within the Trusted Secure IP, and any external access can be shut out to obtain a system with strong security.

55.1 Overview

Table 55.1 summarizes the specifications of the Trusted Secure IP. Figure 55.1 shows a block diagram of the Trusted Secure IP.

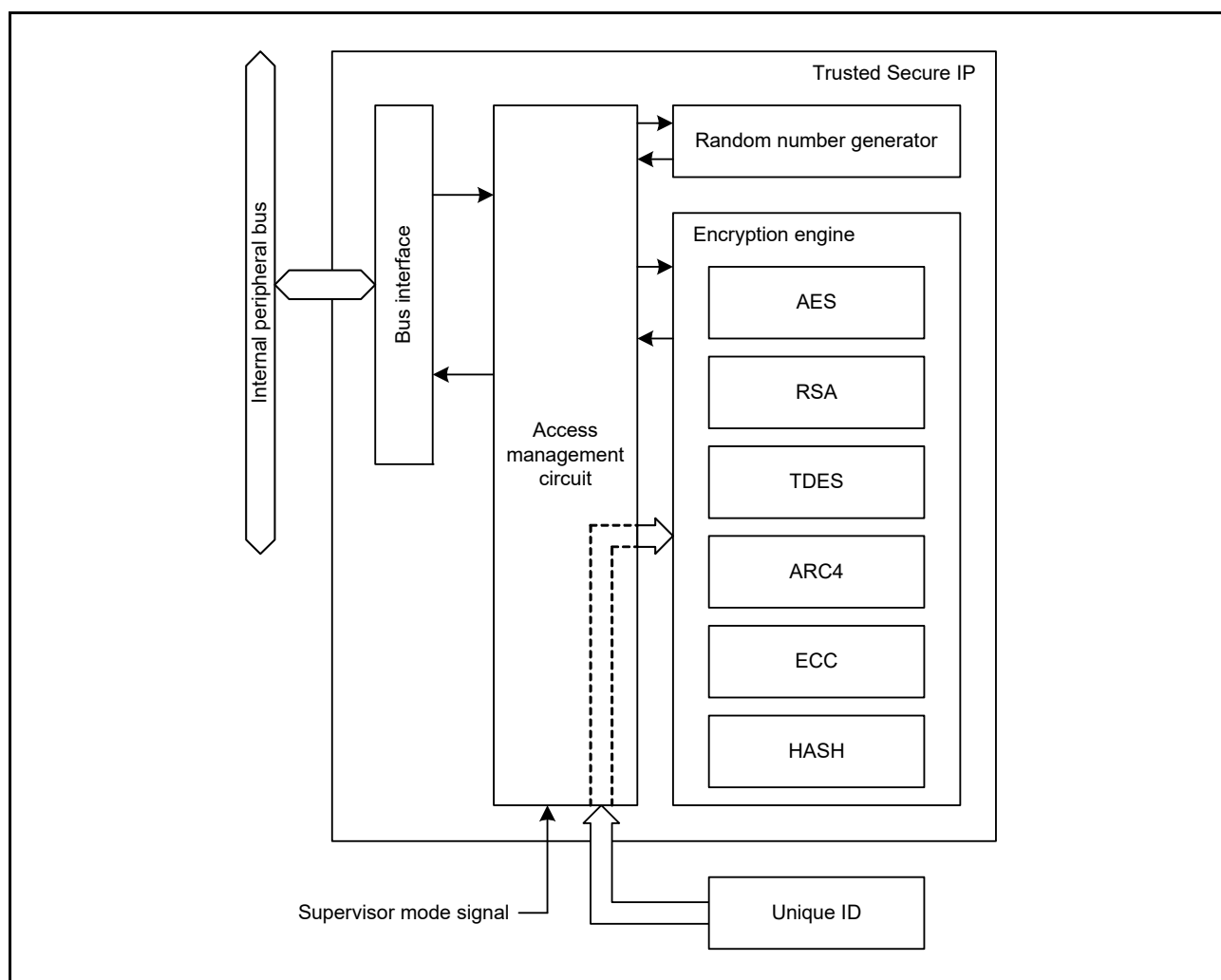
Table 55.1 Specifications of Trusted Secure IP (1/2)

Item	Description
Access control	Access management circuit In case of irregular access to the Trusted Secure IP due to a falsified program or runaway execution of a program, this circuit blocks all subsequent access and stops the output of data from the Trusted Secure IP.
Encryption engine	AES: Compliant with NIST FIPS PUB 197 algorithm - Key sizes: 128, 192, or 256 bits - Block sizes: 128 bits - Block cipher mode of operation - ECB, CBC, CTR: Compliant with NIST SP 800-38A - CMAC: Compliant with NIST SP 800-38B - CCM: Compliant with NIST SP 800-38C - GCM: Compliant with NIST SP 800-38D - XTS: Compliant with NIST SP 800-38E - GCTR - Number of cycles for execution*1 - ECB, CBC, CTR, CMAC, GCTR, XTS: 11 cycles of PCLKB for 128-bit keys, 13 cycles of PCLKB for 192-bit keys, 15 cycles of PCLKB for 256-bit keys - CCM: 22 cycles of PCLKB for 128-bit keys, 26 cycles of PCLKB for 192-bit keys, 30 cycles of PCLKB for 256-bit keys AES-GCM - AES-GCM is realized by combining AES-GCTR and GHASH. RSA - Key sizes: Up to 2048 bits - Block sizes: Up to 2048 bits - Number of cycles for execution: Approximately 1,300,000 cycles of PCLKB when the CRT is used*1 TDES - Key sizes: 56 bits, 2 × 56 bits, or 3 × 56 bits - Block sizes: 64 bits - Block cipher mode of operation: ECB, CBC - Number of cycles for execution*1 16 cycles of PCLKB for 56-bit keys, 32 cycles of PCLKB for 2 × 56-bit keys, 48 cycles of PCLKB for 3 × 56-bit keys ARC4 - Key sizes: 2048 bits - Block sizes: 128 bits - Number of cycles for execution: 16 cycles of PCLKB*1 ECC - Key sizes: Up to 256 bits - Block sizes: 256 bits HASH - Support for SHA1, SHA224/SHA256/MD5, GHASH - Block sizes: 512bits - Number of cycles for execution*1 - SHA1: 80 cycles of PCLKB - SHA224/SHA256/MD5: 64 cycles of PCLKB - GHASH: 9 cycles of PCLKB

Table 55.1 Specifications of Trusted Secure IP (2/2)

Item	Description
Encryption engine	Key management • Keys are only valid within the Trusted Secure IP. • Only key generation information is output from the Trusted Secure IP. • Keys can be regenerated by the input of key generation information to the Trusted Secure IP. Endian • Big or little
Generation of random numbers	32-bit true random number generator • The Trusted Secure IP library can assemble 32-bit true random numbers to generate 128- or 256-bit true random numbers. • The generated 128-bit and 256-bit true random numbers are used as keys in encrypting and decrypting data.
Protection against illicit key copying	• An ID unique to the MCU (unique ID) is accessible from the access management circuit through the dedicated bus. • Combining the unique ID with the key generation information prevents the illicit copying of the key to another MCU.
Supervisor mode	• The supervisor mode signal is connected to the access management circuit and is used to limit control of the Trusted Secure IP module to supervisor mode only.
Interrupt sources	11
Low power consumption	Setting of the module stop state is possible.

Note 1. This does not include the overhead for calling functions of the Trusted Secure IP library.

**Figure 55.1 Trusted Secure IP Block Diagram**

55.2 Operation

55.2.1 Operating Modes and State Transitions

Figure 55.2 shows the state transitions of the Trusted Secure IP.

Use of the Trusted Secure IP security functions is only possible through use of the Trusted Secure IP library provided by Renesas Electronics, in accordance with the state transitions as shown in the figure below.

When irregular access to the Trusted Secure IP (access that violates the defined procedure) due to a falsified program or a program entering runaway execution, etc. is attempted, the access management circuit does not accept any subsequent access and stops the output of any data from the Trusted Secure IP.

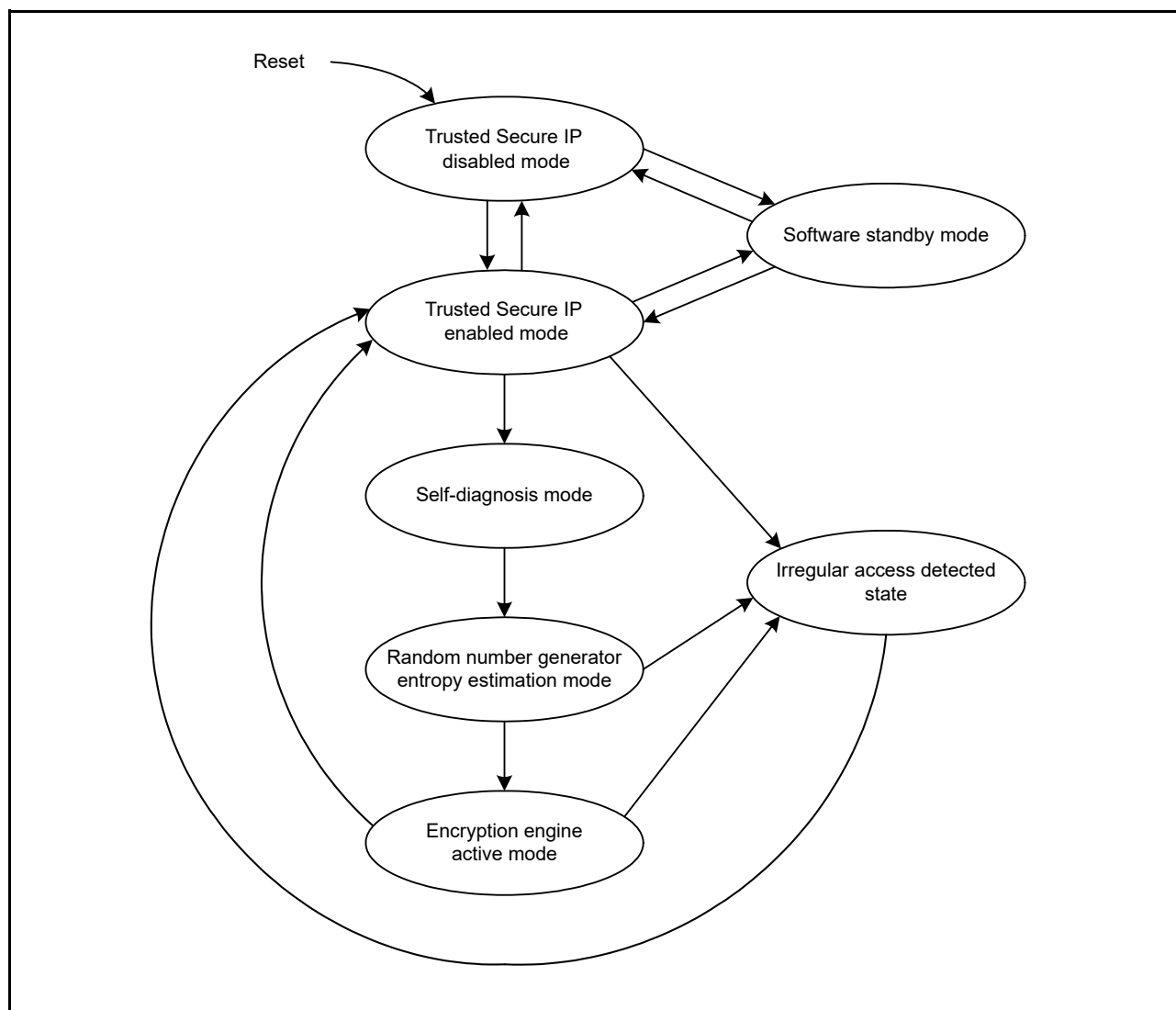


Figure 55.2 Trusted Secure IP Operating Modes and State Transitions

Many of the security functions that the Trusted Secure IP offers are applicable only in the encryption engine active mode. The operations that can be performed in this mode are given below.

- (1) Key Installation
- (2) Encryption and decryption
- (3) Key generation
- (4) Random number generation

55.2.2 Encryption Engine

Figure 55.3 shows processes of the encryption engine integrated in the Trusted Secure IP.

The encryption engine, using the key generation information, performs plaintext to ciphertext encryption and ciphertext to plaintext decryption by hardware.

In no part of the encryption or decryption process, is key data or intermediate data ever exposed outside of the Trusted Secure IP.

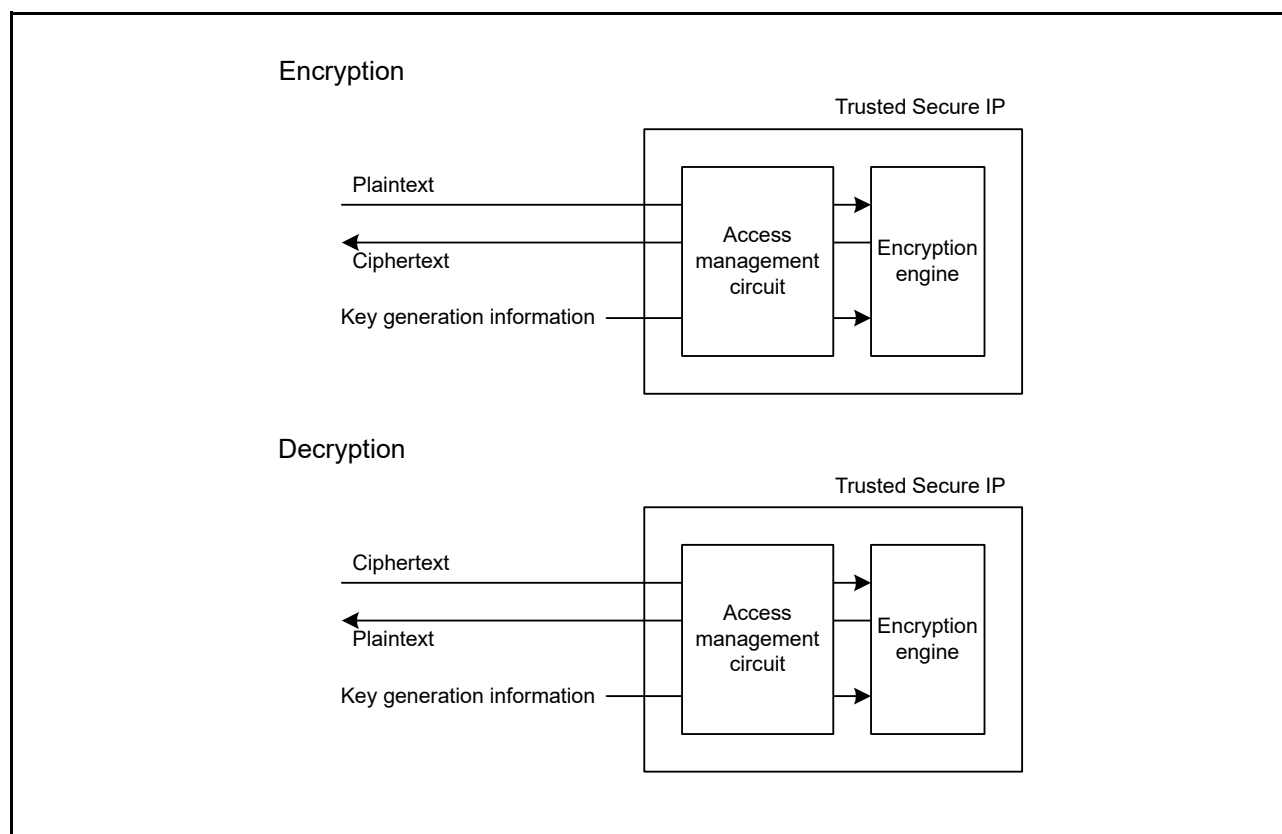


Figure 55.3 Encryption and Decryption processes by Encryption Engine

55.2.3 Key Installation

The key installation is the operation that safely converts the user key to the key generation information and stores it in flash memory. The procedure for installing the key data are given below.

- (1) The user uses the key (Key-2) used for encrypting the user key to encrypt the user key (Key-1) producing eKey-1.
- (2) The user sends the encrypted user key (eKey-1) to the Trusted Secure IP over the serial interface.
- (3) The key generation information of the Key-2 (Index-2) contained in the Trusted Secure IP library is used to recover the Key-2, which is then used to decrypt the user key.
- (4) The user key is converted to user key generation information (Index-1) using the unique ID and a random number, and stored in flash memory.

The installation process and flow chart are given in Figure 55.4 and Figure 55.5, respectively.

Once the key data is installed, the user key generation information (Index-1) can then be used to perform encryption or decryption.

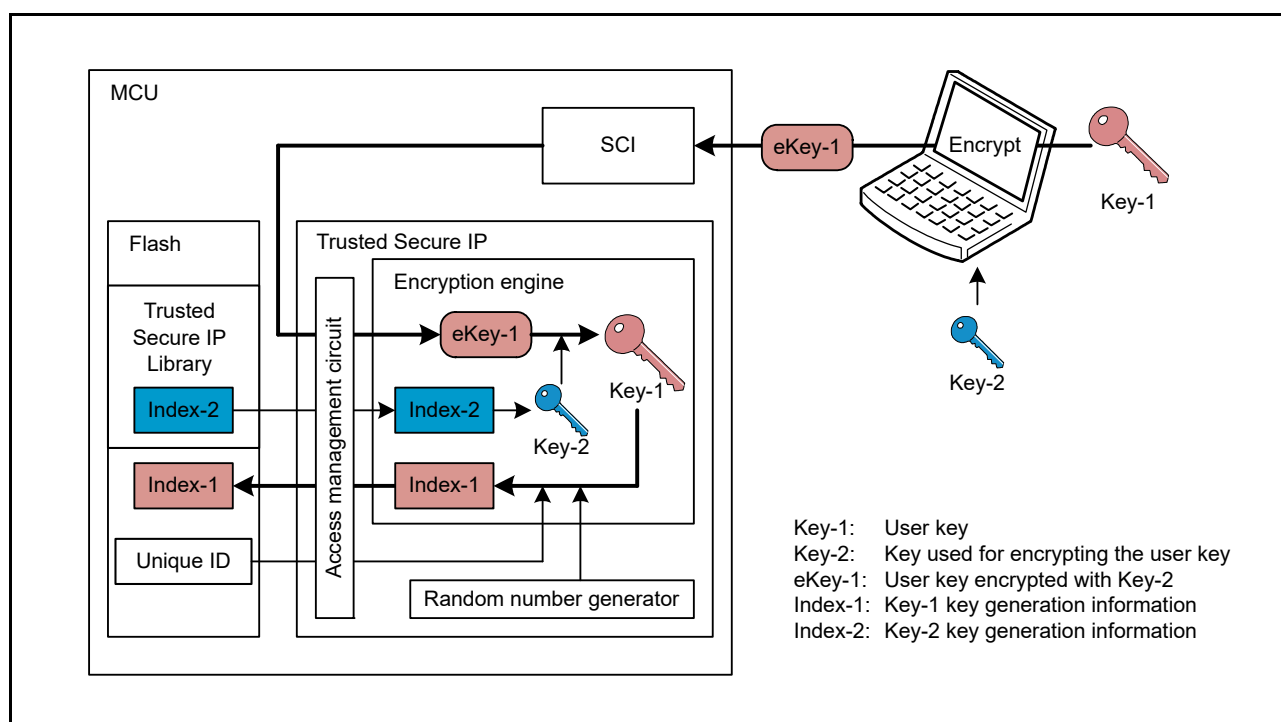


Figure 55.4 Key Installation Process

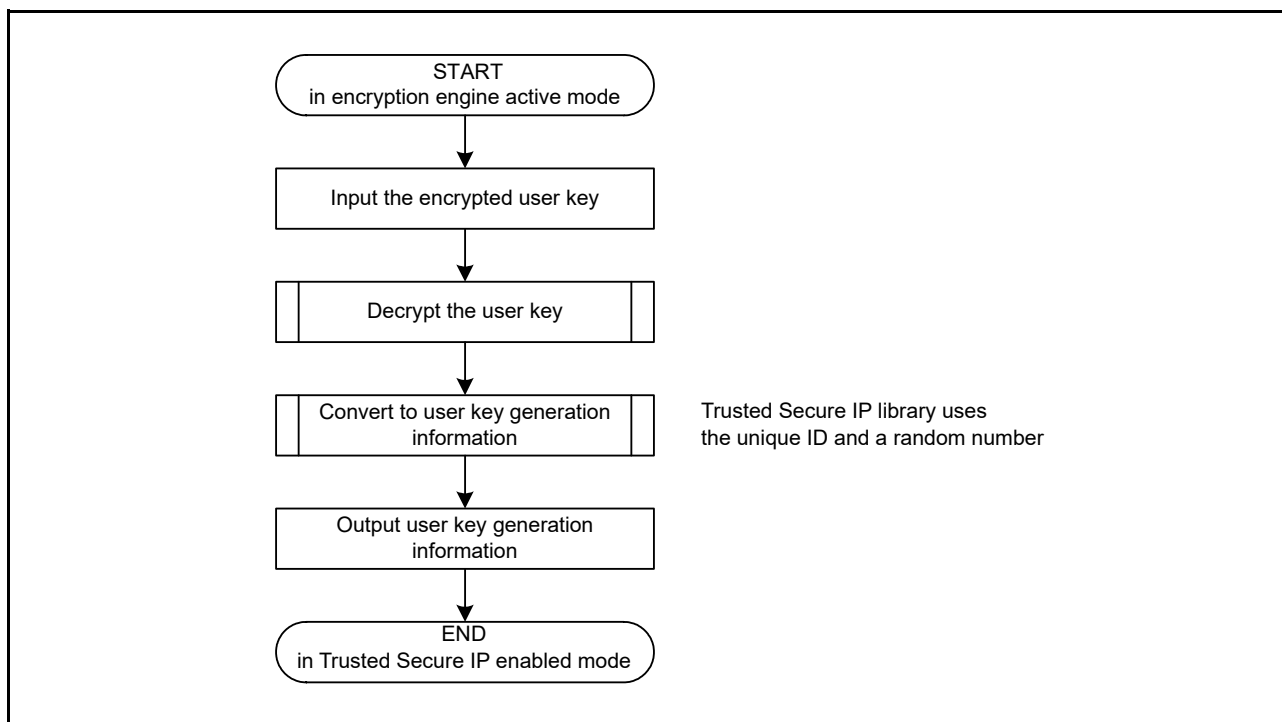


Figure 55.5 Key Installation Flow Chart

55.2.4 Encryption and Decryption

The procedures for encrypting and decrypting data are given below.

- (1) Input the key generation information into the Trusted Secure IP, and recover the key data.
- (2) Input the data to encrypt or decrypt into the Trusted Secure IP. This converts plaintext into ciphertext, and ciphertext into plaintext.
- (3) Read the converted data.

The encryption engine has an input buffer and an output buffer, enabling encryption/decryption to proceed in parallel with data input/output.

Figure 55.6, Figure 55.7, and Figure 55.8 show the timing diagram, encryption flow, and decryption flow, respectively.

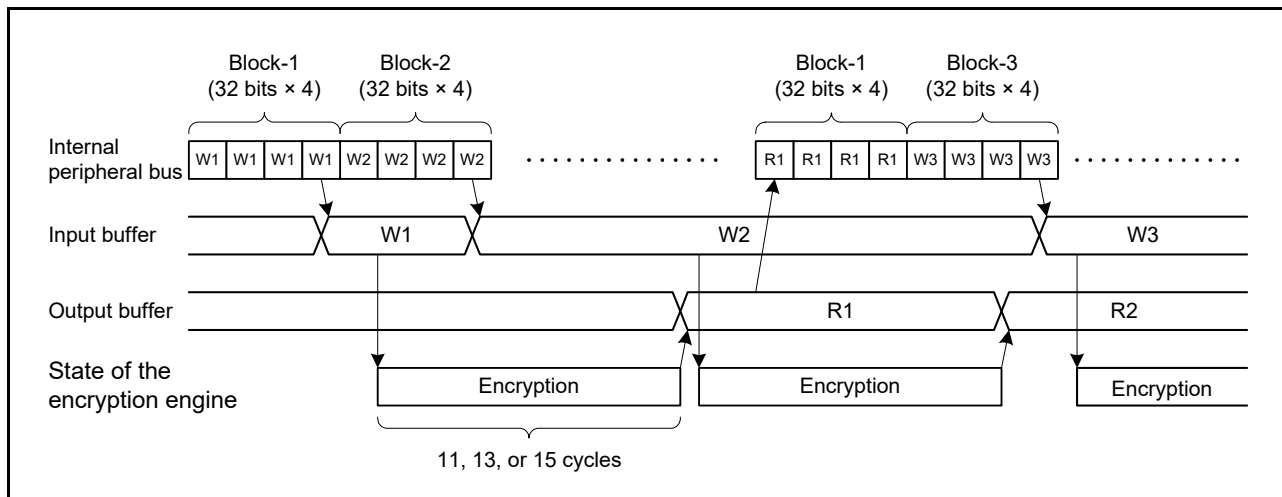
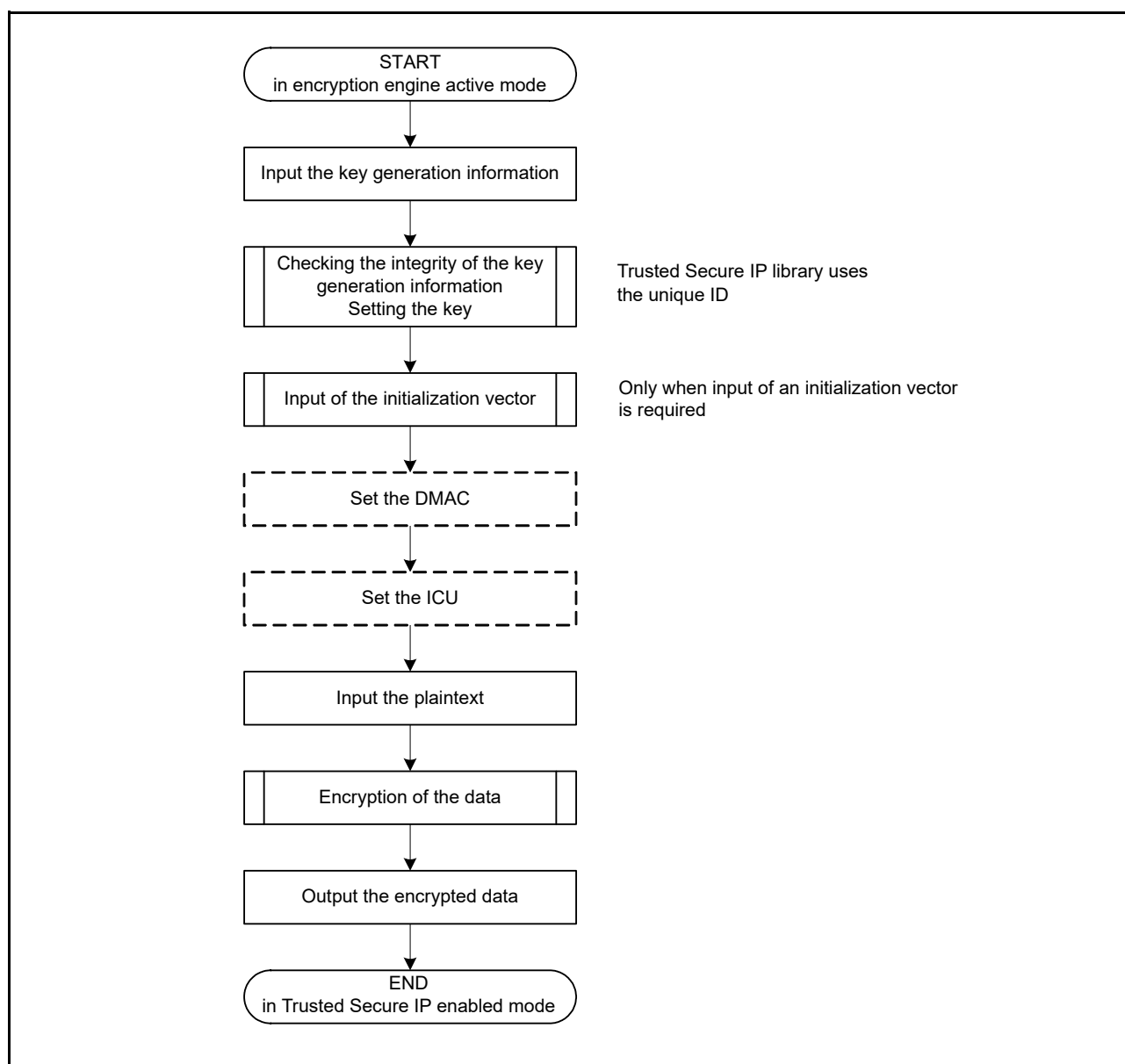
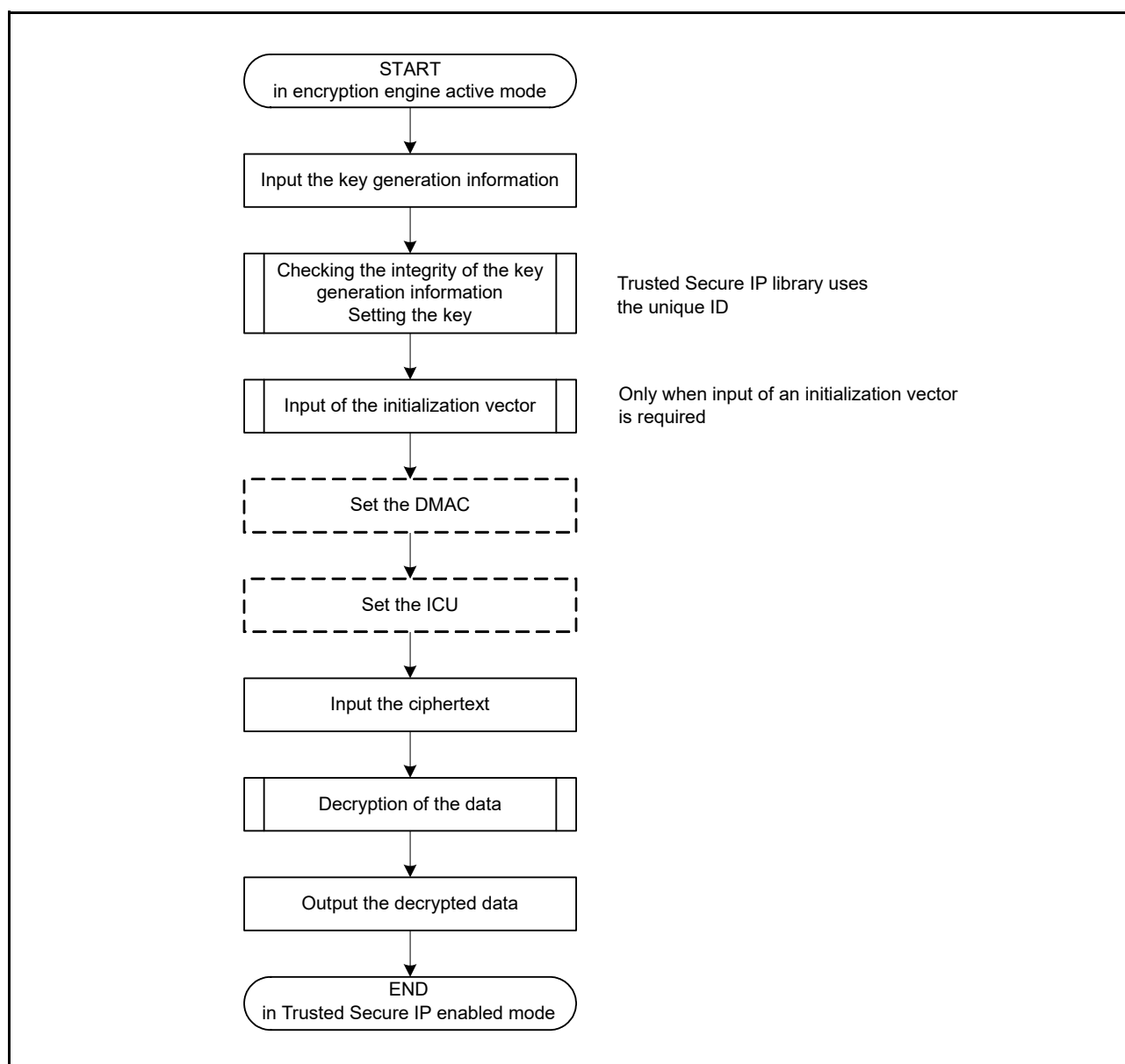


Figure 55.6 Encryption and Decryption Timing Diagram

**Figure 55.7 Encryption Flow Chart**

**Figure 55.8 Decryption Flow Chart**

55.2.5 Generating Key Generation Information (by Using Random Numbers)

Figure 55.9 shows the generating flow for the key generation information by using random numbers.

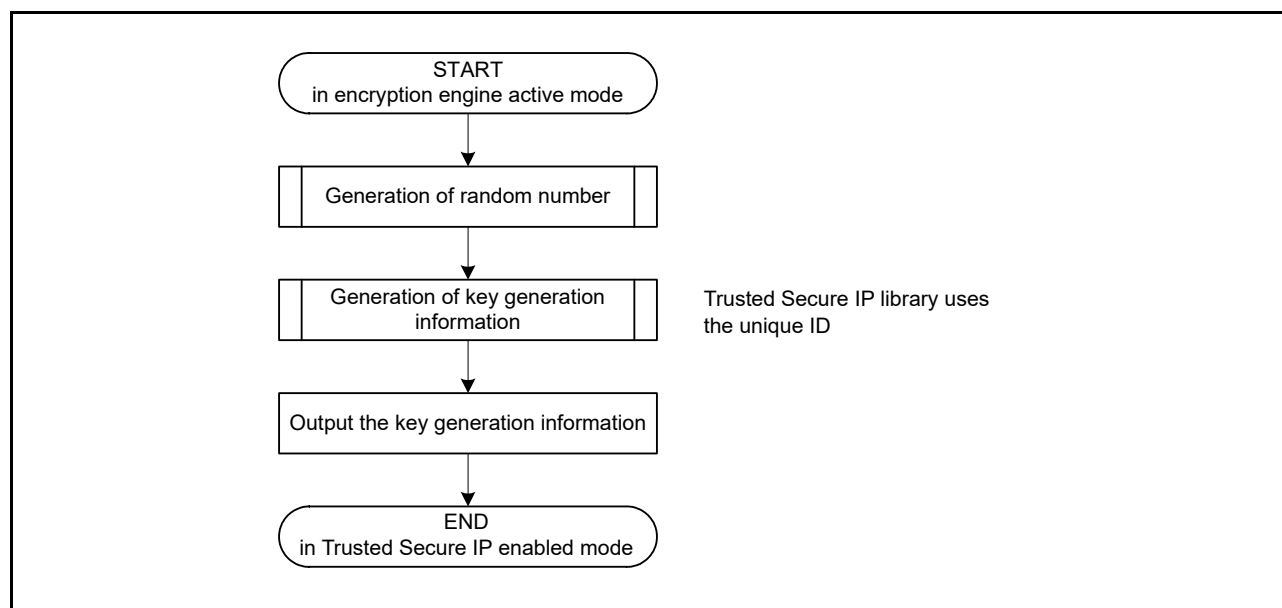


Figure 55.9 Key Generation Information Generating Flow Chart (Using Random Numbers)

55.2.6 Random Number Generation

Figure 55.10 shows the random number generation flow.

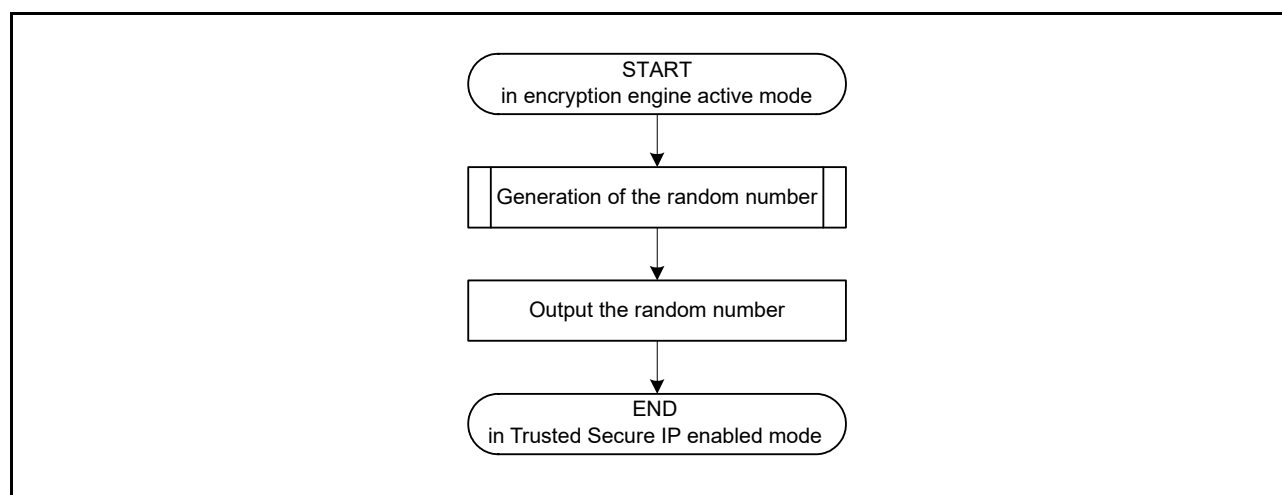


Figure 55.10 Random Number Generation Flow Chart

55.3 Interrupt

Table 55.2 lists the interrupt sources.

Trusted Secure IP library uses interrupts caused by these interrupt sources. Do not set the ICU.IERm.IENj bits corresponding to these interrupt sources to 0.

Table 55.2 Trusted Secure IP Interrupt Sources

Name	Interrupt Source
PROC_BUSY	Procedure completion interrupt
ROMOK	Falsification detection interrupt
LONG_PLG	Calculation completion interrupt
TEST_BUSY	Test busy
WRRDY0	Write ready 0
WRRDY1	Write ready 1
WRRDY4	Write ready 4
RDRDY0	Read ready 0
RDRDY1	Read ready 1
INTEGRATE_WRRDY	Integration write ready
INTEGRATE_RDRDY	Integration read ready

55.4 Usage Notes

55.4.1 Standby Mode

When standby mode is entered while the encryption engine is in processing, proper processing cannot be resumed after standby mode is exited. Standby mode should therefore be entered only after first entering Trusted Secure IP disabled mode or Trusted Secure IP enabled mode.

55.4.2 Setting the Module Stop Function

The module stop control register D (MSTPCRD) enables or disables operation of the Trusted Secure IP. After a reset, the Trusted Secure IP is stopped. After exiting the module stop state, the Trusted Secure IP can be accessed. Refer to **section 11, Low Power Consumption** for details.

55.4.3 Trusted Secure IP Library

Use of the Trusted Secure IP requires the Trusted Secure IP library provided by Renesas Electronics. Please contact our sales office for information regarding the Trusted Secure IP library.